

Application of Deep Reinforcement Learning Algorithms to the Optimal Hedging Problem

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Background

- **Hedging:** Mitigate unwanted risks and prevent large exposures.
- **Greek-Based Approach:** Use Black–Scholes sensitivities (Greeks) to offset risk.
- **Limitations:** Depends on strong model assumptions; may perform poorly under real-world frictions.
- **RL Formulation:** Cast hedging with transaction costs as a sequential decision task and learn dynamic policies via deep RL.

Research Hypothesis and Evaluation

Research Hypothesis

A data-driven reinforcement learning approach to hedging a derivative portfolio can be used to reduce the risk of tail events.

- **Reduce Tail Risk Metrics:**
 - Lower Value at Risk (VaR)
 - Lower Conditional Value at Risk (CVaR)
- **Ensure Model-Agnostic Robustness:**
 - Strategy performance independent of specific simulator dynamics
 - Validated across multiple stochastic market models

General Setup

- **Objective:** Hedge a liability portfolio (an ATM European call, one-month maturity) against tail-loss events by trading the underlying stock daily until expiration.
- **Transaction Costs:** Proportional cost rate $\lambda_i > 0$ for asset i .

$$\Delta H_{t+1}^i = H_{t+1}^i - H_t^i, \quad C_t(H) = \sum_{i=0}^N \lambda_i |\Delta H_{t+1}^i| S_t^i.$$

- **Portfolio Value:**

$$\text{Gross value: } V_t(H) = \sum_{i=0}^N H_t^i S_t^i.$$

(Net/self-financing definition adjusted to account for $C_t(H)$ in subsequent analysis.)

- **Rebalancing:** Daily adjustments incur costs $C_t(H)$; standard self-financing condition is modified accordingly.

Trading Dynamics

- At each time $t = 0, \dots, T - 1$:
 - Observe market state.
 - Choose new hedge portfolio H_{t+1} .
 - Trade $\Delta H_{t+1} = H_{t+1} - H_t$, paying cost

$$C_t(H) = \sum_{i=0}^N \lambda_i |\Delta H_{t+1}^i| S_t^i.$$

- Initial portfolio value $V_0(H) = 0$ (borrow/lend at $r = 0$).
- At maturity $t = T$:

$$V_T(H) = \sum_{i=0}^N H_T^i S_T^i \longrightarrow \text{liquidate \& pay } X.$$

- Profit/Loss:

$$\text{PnL}(H) = V_T(H) - X(\omega).$$

Reinforcement Learning Formulation

In our setting, we employ a POMDP

$$(\mathcal{S}, \mathcal{A}, \Omega, \mathcal{T}, \mathcal{O}, \mathcal{R})$$

to model the interaction between the hedging agent and the financial market.

Action Space

- **Action Space $\mathcal{A}$:** At each time t , the agent selects an action corresponding to the new weights in the hedging portfolio H_{t+1} .

$$a_t \in \mathcal{A},$$

- Actions encode buy/sell decisions across instruments (e.g. fractional positions in $[0, 1]$) and determine ΔH_{t+1} .

State Space and State Transition

Hidden State Space $\mathcal{S}$: Represents the true, unobservable factors driving asset dynamics.

State Transition Kernel $\mathcal{T}$: Governs the stochastic evolution of the hidden state. In our implementation,

$$s_{t+1} \sim \mathcal{T}(s_{t+1} \mid s_t, a_t)$$

is realized via Heston model dynamics.

Heston Stochastic Volatility Model:

- Models both asset price S_t and instantaneous variance v_t .
- Captures mean-reversion of variance and price–variance correlation.
- Simulated paths serve as “real” market data for training.

Heston Simulator

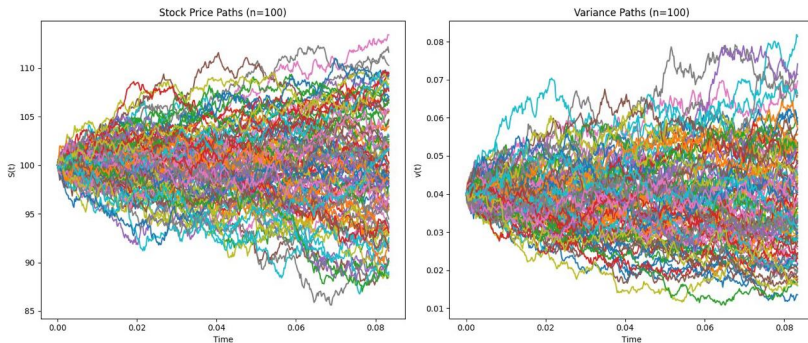


Figure: Heston model simulation paths for price and variance

S_0	v_0	θ	σ	κ	r	q
100	0.04	0.04	0.28	1	0.00	0.00

Observation Space and Observation Model

- **Observation Space Ω :** At each time t , the agent observes a vector

$$o_t = (\tilde{S}_t, \tilde{v}_t, a_t, \tau_t, \widetilde{\text{PnL}}_t, m_t),$$

- Where all components except m_t are normalized to lie in a bounded interval (e.g. $[-1, 1]$) to stabilize training and modeling.

- **Observation Model $\mathcal{O}$:**

$$o_{t+1} \sim \mathcal{O}(o_{t+1} \mid s_{t+1}, a_t)$$

- Observations (including realized variance) depend on s_{t+1} and a_t , with no lookahead.
- Heston parameters $(\kappa, \theta, \sigma, \rho)$ are unknown to the agent.

Reward Function

- **Intermediate Reward** ($0 \leq t < T$):

$$\begin{aligned} R_t(s_t, a_t) &= \widehat{\text{PnL}}_t - \widehat{\text{PnL}}_{t-1} \\ &= [-\hat{C}_t + V_t(H)] - [-\hat{C}_{t-1} + V_{t-1}(H)], \end{aligned}$$

- **Terminal Reward** ($t = T$):

$$R_T(\text{PnL}_T) = \alpha \exp\left(-\frac{\text{PnL}_T^2}{2\sigma^2}\right),$$

which sharply penalizes large hedging errors.

- Here, α controls the peak reward at zero error, and σ defines the acceptable error width.

PnL Estimation & Reward Shaping

- **ATM Call Price Heuristic:** Following quick approximation for the price of an at-the-money (ATM) call option:

$$\hat{C}_{\text{ATM}} \approx 0.4 S_t \sigma \sqrt{\tau}.$$

- **Piecewise Approximation:**

$$\hat{C}_t = \begin{cases} \max(S_t - K, 0), & \tau = 0, \\ (S_t - K e^{-r\tau}) + 0.4 S_t \sqrt{V_t \tau}, & S_t > K, \\ 0.4 S_t \sqrt{V_t \tau} \frac{S_t}{K}, & S_t \leq K. \end{cases}$$

Terminal Reward Function

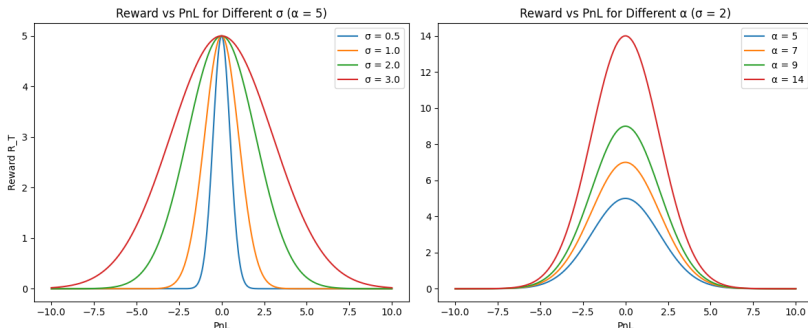


Figure: Terminal reward plot for a grid of parameters

Policy

- A policy maps the agent's information history $h_t = (o_0, a_0, o_1, a_1, \dots, o_t)$ to an action distribution:

$$\pi_t(a_t \mid h_t)$$

- A policy decides the action a_t based on past observations and actions.
- In deep RL, a neural network with parameters θ outputs either:
 - A probability distribution over actions, or
 - A deterministic action in $[0, 1]$.
- The network is trained to maximize expected cumulative reward.

Experimental Setup and Evaluation Protocol

- **Hedged Instrument:** European call option, strike $K = 100$, one-month maturity.
- **Transaction Cost:** Proportional cost $\lambda = 0.001$ per traded position.
- **Time Discretization:** One-month horizon divided into 400 trading steps.
- **Training:** Each model trained for 40,000 time-steps (no data leakage).
- **Evaluation:**
 - Out-of-sample test set of 10,000 simulated price paths.
 - Performance assessed using Value-at-Risk and Conditional Value-at-Risk:

$$\text{VaR}_\alpha = -\inf\{x: F(x) \geq \alpha\}, \quad \alpha \in [0, 1],$$

$$\text{CVaR}_\alpha = -\mathbb{E}[X \mid X \leq -\text{VaR}_\alpha].$$

TD3 & DDPG Policy Results

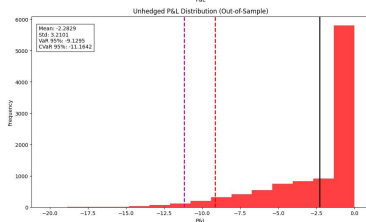
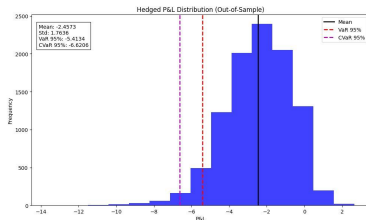


Figure: TD3: $\alpha = 5, \sigma = 0.5$

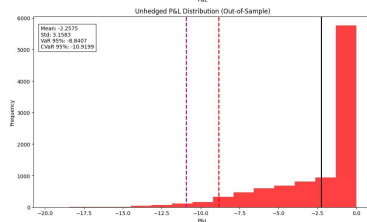
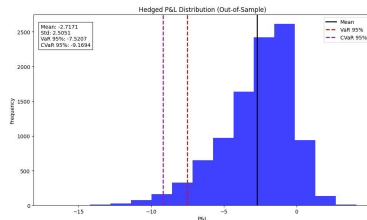


Figure: DDPG: $\alpha = 7, \sigma = 0.5$

SAC Policy Results

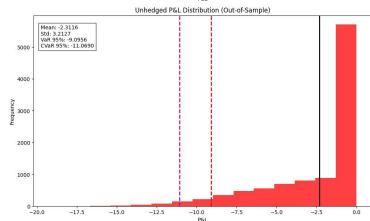
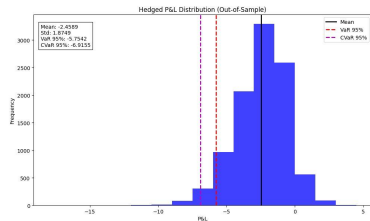


Figure: $\alpha = 7, \sigma = 0.5$

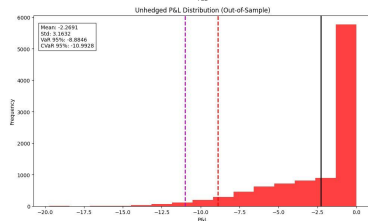
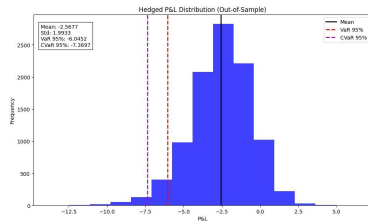


Figure: $\alpha = 9, \sigma = 1$

Robustness Across Market Models

The same framework was applied under the Black–Scholes–Merton model, where the stock price follows geometric Brownian motion with constant volatility.

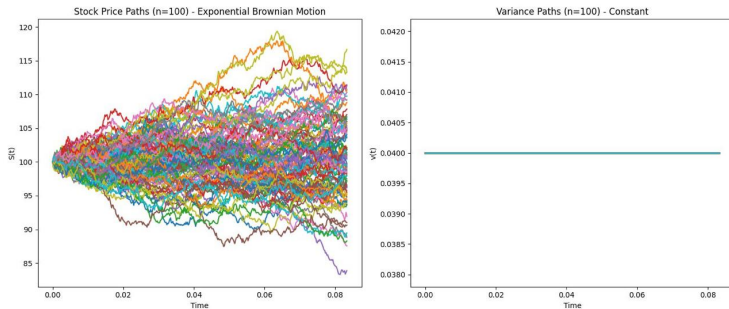


Figure: One hundred simulated paths of the BSM model for stock price (left) and constant variance (right)

P&L Distributions under BSM

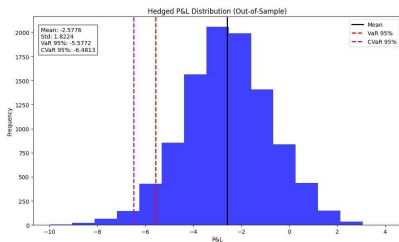


Figure: TD3 P&L distribution under the BSM model with $\alpha = 5$ and $\sigma = 1$.

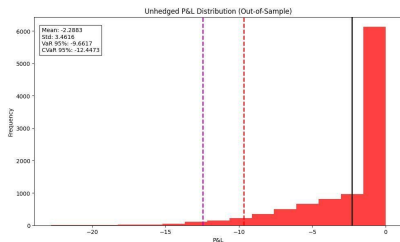


Figure: Unhedged P&L distribution under the BSM model

Future Work

Future improvements fall into two main areas:

- **Enhanced Market Simulator:**

- Extend to multi-asset, multi-product environments.
- Capture joint dynamics of equities, options, and other derivatives.

- **Refined Agent Framework:**

- Explicitly model agent's risk preferences.
- Enable direct control over acceptable tail outcomes.
- Support targeted hedging behavior.

Shaping the reward to reflect risk preferences could yield an indifference price, accounting for both market frictions and the agent's tolerance.

Conclusions

Takeaway Message

Model-free, data-driven reinforcement learning approach can yield practically valuable hedging strategies that respect transaction costs, leverage constraints, and partial observability

- Formulation of the financial problem utilizing POMDP framework
- Creation of custom hedging environment
- Successful Utilization of Deep Reinforcement Learning Algorithms for Optimal Hedging

Questions & Discussion

Thank you for your attention!

Any questions?

All of the code for this project is available at
<https://github.com/platonbarunin/diploma>